

HDOC - Day 14 Activity

C Programming Assignment

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Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the two operations: (1) check whether all digits are unique, (2) find the frequency of each digit from 0 to 9. Prepare the algorithm and test cases, then compile, execute & test the program.

Algorithm

1. Start.
2. Input a positive integer number.
3. If number ≤ 0 , display an invalid-input message and stop.
4. Display the menu: 1 - Check unique digits, 2 - Find digit frequencies.
5. Input the user's choice. Set temp = number and initialize freq[10] to zero.
6. Use switch(choice).
7. Case 1: Repeatedly take digit = temp % 10. If freq[digit] is already greater than 0, the number is Not Unique. Otherwise increment freq[digit] and remove the last digit using temp = temp / 10. If no digit repeats, display Unique.
8. Case 2: Repeatedly take digit = temp % 10, increment freq[digit], and remove the last digit. Then display freq_0 through freq_9.
9. Default: Display Invalid choice if the user enters anything other than 1 or 2.
10. Stop.

Test Case Table

Test Case	Input(number)	Choice	Expected Output	Result
1	12345	1	Unique	Pass
2	12234	1	Not Unique	Pass
3	112333	2	0:0, 1:2, 2:1,3:3; others 0	Pass
4	987654	2	4-9: 1 each; 0-3: 0	Pass
5	123	3	Invalid choice	Pass
6	-25	1	Invalid input	Pass

C Program

```
#include <stdio.h>

int main(void) {
    long long number, temp;
    int choice, digit, freq[10] = {0};
    int unique = 1;
    printf("Enter a positive integer: ");
    scanf("%lld", &number);
    if (number <= 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 0;
    }
    printf("\n1. Check whether all digits are unique\n");
    printf("2. Find frequency of each digit (0 to 9)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    temp = number;
    switch (choice) {
        case 1:
            while (temp > 0) {
                digit = temp % 10;
                if (freq[digit] > 0) {
                    unique = 0;
                    break;
                }
            }
        case 2:
            while (temp > 0) {
                digit = temp % 10;
                freq[digit]++;
                temp /= 10;
            }
            for (digit = 0; digit < 10; digit++) {
                printf("%d: %d\n", digit, freq[digit]);
            }
            break;
        default:
            printf("Invalid choice\n");
            break;
    }
    if (unique) {
        printf("All digits are unique\n");
    } else {
        printf("Not Unique\n");
    }
}
```

```
freq[digit]++;
temp /= 10;
}
if (unique)
printf("Output: Unique\n");
else
printf("Output: Not Unique\n");
break;
case 2:
while (temp > 0) {
digit = temp % 10;
freq[digit]++;
temp /= 10;
}
printf("Digit frequencies:\n");
for (int i = 0; i <= 9; i++)
printf("freq_%d = %d\n", i, freq[i]);
break;
default:
printf("Invalid choice. Please enter 1 or 2.\n");
}
return 0;
}
```

Compilation , Testing & Output

```
Session Actions Edit View Help
(kali@kali)-[~]
$ vi number.c
(kali@kali)-[~]
$ cc number.c
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

1. Check whether all digits are unique
2. Find frequency of each digit (0 to 9)
Enter your choice: 1
Output: Unique

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12234

1. Check whether all digits are unique
2. Find frequency of each digit (0 to 9)
Enter your choice: 1
Output: Not Unique

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 112333

1. Check whether all digits are unique
2. Find frequency of each digit (0 to 9)
Enter your choice: 2
Digit frequencies:
freq_0 = 0
freq_1 = 2
freq_2 = 1
freq_3 = 3
freq_4 = 0
freq_5 = 0
freq_6 = 0
freq_7 = 0
freq_8 = 0
freq_9 = 0
```

```
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 987654

1. Check whether all digits are unique
2. Find frequency of each digit (0 to 9)
Enter your choice: 2
Digit frequencies:
freq_0 = 0
freq_1 = 0
freq_2 = 0
freq_3 = 0
freq_4 = 1
freq_5 = 1
freq_6 = 1
freq_7 = 1
freq_8 = 1
freq_9 = 1

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 123

1. Check whether all digits are unique
2. Find frequency of each digit (0 to 9)
Enter your choice: 3
Invalid choice. Please enter 1 or 2.

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: -25
Invalid input. Please enter a positive integer.

(kali@kali)-[~]
$
```